

BusBuddy Autonomous Engineering Backlog

Full-Codebase Production Readiness and Accessibility Audit

Chief Accessibility and Mobile Systems Architecture Review

20 September 2026

Audit basis and repository reconciliation

This audit treats the public `main` branch of `sricharan24-byte/HELEN`, retrieved on 20 September 2026, as authoritative. Source was inspected across the available `lib/`, `test/`, `android/`, and `docs/` paths. Direct checkout and local execution were blocked by the workspace network proxy, so the reported test count was not independently rerun. Every issue below is tied to code visible in the public repository; test status is therefore “reported”, not independently verified.

The inspected branch does not match the supplied completion inventory:

- The README reports 214 tests, not 236.
- `docs/adr/ADR-002-phase2-accessibility-and-safety-architecture.md` is not available at the stated public path.
- The fare engine is at `lib/domain/ticketing/entities/fare_engine.dart`, while production ticket flows continue through legacy `double` models and hard-coded card fares.
- Semantic colors are in `lib/core/tokens/app_semantic_colors.dart`; the light/dark primary blue with white text measures approximately 4.10:1, below WCAG AA for normal text, and high-contrast error red on the high-contrast background measures approximately 5.15:1, below the claimed 7:1 AAA target.
- The floating overlay does not contain the claimed `BlockSemantics` barrier or accessible corner-reposition actions.
- Android voice capture returns a delayed hard-coded phrase and Android speech playback completes without producing audio. The full voice implementation is web-only.

These are release-relevant source/documentation discrepancies, not naming preferences.

1 Section A: Production Readiness Scorecard

Area	Score	Source-backed rationale
Domain Isolation & Currency Integrity	4/10	Pure-Dart domain types and integer-paise Fare exist, but live repositories and UI use duplicate legacy models, double fare fields, hard-coded bus fares, and a second fare formula. FareEngine is not the production authority end-to-end.
Async Lifecycle & Teardown Safety	4/10	The locator centralizes transport instances, but teardown does not await StreamController.close() , app-scoped ChangeNotifiers and the floating/audio singletons are not locator-owned/disposed, and late async work lacks generation cancellation.
WCAG 2.2 / Astra Accessibility Compliance	4/10	There are useful semantics, map text alternatives, and a 48dp token. However the app replaces nonlinear platform scaling with a linear multiplier, permits downscaling, contains sub-48dp controls, duplicate compound semantics, fixed-height layouts, hidden-step announcements, incomplete modal isolation, and unverified contrast claims.
Multimodal Voice & Audio Architecture	3/10	Web PCM/TTS/STT paths exist, but Android uses simulated speech, the announcement coordinator is not integrated with assistant audio, reconnect/backoff and lifecycle pause are absent, and an emergency tool reports success before an emergency action occurs.
Fault Tolerance & Offline Degradation	3/10	Local transport fixtures provide a useful fallback, but OSRM has no timeout or status validation, map tiles expose no deterministic recovery state, live telemetry has no staleness/out-of-order policy, and most failures are silently replaced with apparently live values.
Production Release Hardening	2/10	The Android release build uses the debug signing configuration and placeholder application ID; shrinking is disabled, permissions exceed implemented native behavior, ticket/payment/QR flows are demonstrative but visually presented as real, and release evidence is not CI-generated.

Overall production readiness: 3.3/10. The repository is a capable prototype, but it is not ready for a production transit, payment, emergency, or accessibility claim until all P0 tasks and the P1 accessibility/fault-tolerance tasks are closed.

2 Section B: Prioritized Production Implementation Tasks

P0 — Release blockers

BUS-P0-01: Make FareEngine the sole end-to-end fare authority

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gate 1 (deterministic domain authority); WCAG 2.2 SC 3.3.4

Error Prevention (Legal, Financial, Data)

Target Files & Symbols: `lib/domain/ticketing/entities/fare.dart` `Fare`; `lib/domain/ticketing/entities/fare_engine.dart` `FareEngine`; `lib/data/models/ticket_model.dart` `Ticket.calculateFare`, `TicketRepository.bookTicket`; `lib/features/tickets/booking_checkout_dialog.dart` `_issueTicket`, `_FareSummary`; `lib/features/tickets/ticket_booking_suite_page.dart` `_buildBusCard`

Observed Defect / Gap: The domain stores paise but accepts a `double` rupee constructor. The legacy ticket model stores `doublefare` and recalculates concession independently. Checkout receives a hard-coded base fare from visual bus cards (Rs. 25/Rs. 30), while the displayed stop count is artificially clamped to 1–3. Neither booking nor repository persistence calls `FareEngine.calculateByStopCount`. Thus the supposedly authoritative engine can disagree with the issued ticket and receipt.

Required Production Solution: Introduce one integer-only `Money/FareQuote` contract containing base paise, discount paise, final paise, passenger type, hop count, and rule version. Remove production `double` fare fields/constructors and duplicate formulas. Resolve ordered boarding/alighting stop occurrences first, call `FareEngine` once, pass the immutable quote through confirmation and booking, and persist the exact quote. Format rupees only at presentation boundaries.

Test Specification: Add `test/integration/ticket_fare_authority_test.dart`. Book general/student/senior tickets at 1, 3, 4, 5, and 6 hops in both directions; assert exact paise values 1500/2000/2500 and 900/1200/1500. Assert `origin=destination` and unordered stops return `ValidationFailure`; assert displayed checkout total, stored ticket, QR payload, and receipt all equal the same quote. Add a source guard that fails on a second production fare formula or `doublefare`.

Exit Criteria: Every issued ticket is derived from one `FareQuote`; no production ticket path contains a hard-coded fare or floating-point currency; boundary and end-to-end tests pass.

BUS-P0-02: Enforce a durable ticket/payment state machine and replay protection

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gates 1 and 12; WCAG 2.2 SC 3.3.4 and 3.3.6 Accessible Authentication

Target Files & Symbols: `lib/domain/ticketing/entities/ticket.dart` `Ticket`, `TicketStatus`; `lib/data/models/ticket_model.dart` `Ticket`, `TicketRepository`; `lib/features/tickets/ticket_controller.dart` `bookTicket`, `cancelTicket`; `lib/features/tickets/booking_checkout_dialog.dart` `_issueTicket`; ticket detail/QR validation surfaces

Observed Defect / Gap: Tickets are created directly as active, cancellation is an unrestricted `copyWith`, there is no activation/validation/refund transition contract, no persisted ledger, no idempotency key, and IDs combine milliseconds with only 1,000 random values. Checkout labels a button “Pay” but immediately issues a ticket without payment authorization. The QR payload is predictable unsigned text, so it can be forged or replayed; its embedded expiry can differ from the ticket expiry when a past travel date triggers the fallback validity.

Required Production Solution: Define legal transitions `quoted->paymentPending->issued->active->used/expired` with cancellation/refund terminal branches. Make transition methods return `Result`; record immutable transition timestamps and transaction/idempotency IDs; reject repeated call-backs and terminal-state mutation. Persist tickets atomically. Use server-issued opaque signed tokens for real validation, or explicitly label and feature-gate the entire flow as a non-production demo. Correct QR expiry to serialize the finalized ticket expiry.

Test Specification: Add `test/domain/ticket_state_machine_test.dart` and `test/integration/ticket_replay_test.dart`. Exercise every legal/illegal transition, duplicate payment callback, concurrent cancellation/use, clock expiry, app restart, duplicate ID attempt, forged QR, replayed QR, and past travel date. Assert one ledger entry, one charge, immutable amount/rule version, and stable terminal state.

Exit Criteria: Illegal/replayed transitions are impossible through public APIs; tickets survive process restart; payment and QR behavior is either production-authenticated or unambiguously disabled/labeled as demo.

BUS-P0-03: Preserve the platform `TextScaler` without downscaling or linearization

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gates 3 and 10; WCAG 2.2 SC 1.4.4 Resize Text and 1.4.10 Reflow

Target Files & Symbols: `lib/main.dart` `BusBuddyApp.build`; `lib/core/settings/app_settings_controller.dart` `AppTextSize`, `AppSettingsController.textScale`

Observed Defect / Gap: `MediaQuery.textScaler.scale(1.0)` is multiplied by an app preference and replaced with `TextScaler.linear`. This discards platform nonlinear scaling behavior and allows `small=0.85` to reduce a user's chosen accessibility size. It therefore does not satisfy the claimed uncapped platform-scaler contract.

Required Production Solution: Pass the platform `TextScaler` through unchanged by default. If an explicit in-app enlargement feature is retained, compose an enlargement-only transform without reducing platform output and document/test the semantics at multiple font sizes; never reconstruct the platform scaler from `scale(1.0)`. Deprecate “Small” for accessibility text, or make it affect only app chrome spacing rather than text.

Test Specification: Add `test/all/platform_text_scaler_test.dart`. Inject nonlinear fake scalers and system factors equivalent to 100%, 200%, and 300%; assert platform output is never reduced, nonlinear values remain nonlinear, and all primary routes render at 320dp without overflow exceptions or unreachable actions.

Exit Criteria: No code replaces the platform scaler with a linear approximation or reduces system-selected text; 320dp/300% golden and interaction tests pass.

BUS-P0-04: Make the composition root own and await all app-scoped teardown

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gates 2 and 9; lifecycle reliability prerequisite

Target Files & Symbols: `lib/core/di/service_locator.dart` `AppServiceLocator`, `resetForTesting`; `lib/data/repositories/transport_repository.dart` `dispose`; `lib/data/datasources/live_bus_movement_engine.dart` `dispose`, `_refreshPath`; `lib/features/ai_assistant/floating_assistant_controller.dart` `instance`, `reset`; `lib/features/ai_assistant/audio_speech_engine.dart` `dispose`

Observed Defect / Gap: The repository's async `dispose` calls a void movement-engine `dispose`; the engine invokes `StreamController.close()` without awaiting its returned future. Locator reset disposes only repositories, not its `TicketController/JourneyController`; the floating assistant/audio/session

singleton sits outside the locator. In-flight OSRM/WebSocket callbacks have no generation token and may complete after reset.

Required Production Solution: Create an `AsyncDisposable` contract. Make engine, repositories, controllers, assistant session/audio, and settings ownership explicit in one composition root. Await timer/subscription cancellation and controller close; invalidate late work with a monotonically increasing generation/cancellation token; make start/stop/dispose idempotent; and prevent lazy recreation while reset is running.

Test Specification: Extend `test/core/di/service_locator_lifecycle_test.dart` with concurrent first access, double dispose, reset during an OSRM future, reset during a queued timer tick, 100 create/reset cycles, and stale callback completion. Assert zero active timers/listeners, closed streams, disposed notifiers, no post-dispose notification, and one instance per generation.

Exit Criteria: A single awaited reset leaves no live timer, stream, socket, audio callback, listener, or app-scoped notifier; stress tests are deterministic and green.

BUS-P0-05: Prevent Gemini tools from claiming unexecuted emergency or financial actions

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gate 12; WCAG 2.2 SC 3.3.4; safety-critical truthful feedback

Target Files & Symbols: `lib/features/ai_assistant/gemini_live_session.dart` `_executeToolCall`; `lib/features/ai_assistant/floating_assistant_controller.dart` `_ensureSession`, `_handlePrompt`, `onAction`; safety and ticket action handlers

Observed Defect / Gap: `emergency_sos` returns a successful tool response stating that SOS was activated, but the session only records an action type. The controller later adds a button; it does not execute an SOS broadcast. The model may therefore tell a blind user that emergency help was activated when nothing happened. The same tool framework can represent booking/sharing actions as successful before confirmation or persistence.

Required Production Solution: Split tools into read-only queries and confirmable commands. Commands return `confirmationRequired` until the user activates a labeled non-voice/voice confirmation; execute through a typed command gateway; return success only after the repository/OS operation acknowledges it. Persist an audit event and expose visible, spoken, and haptic success/failure. Emergency failure must retain a direct call/SMS fallback.

Test Specification: Add `test/features/ai_assistant/tool_command_safety_test.dart`. Simulate SOS, share location, and ticket commands; assert no action before confirmation, cancellation is honored, duplicate model calls are idempotent, repository failure returns failure language, and the model never receives a success payload before actual completion.

Exit Criteria: No assistant response can claim an emergency, sharing, booking, or payment action succeeded unless an authenticated application command completed.

BUS-P0-06: Replace simulated Android voice behavior or production-gate it

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Astra Gates 6, 9, and 12; WCAG 2.2 SC 2.1.1 Keyboard and 4.1.3 Status Messages

Target Files & Symbols: `lib/features/ai_assistant/audio_speech_engine.dart` `startListening`, `speak`, `playPcm24k`; `lib/features/ai_assistant/gemini_live_session.dart`; `lib/features/ai_assistant/gemini_live_screen.dart`; Android platform integration and `pubspec.yaml`

Observed Defect / Gap: On non-web platforms `startListening` waits and returns the hard-coded sentence “Where is my bus?”; `speak` waits without speech; PCM playback is discarded after a delay. The Android-first app therefore presents voice controls that do not capture or speak real content.

Required Production Solution: Implement real Android STT/TTS/PCM playback with permission and audio-focus handling, or hide/disable the live controls behind a capability check with an explicit “voice unavailable” explanation and complete keyboard/touch fallback. Do not ship simulated input under a microphone UI. Add platform audio session interruption handling and release resources on background/dispose.

Test Specification: Add Android integration tests for granted/denied/permanently-denied microphone permission, recognized transcript propagation, TTS completion/interruption, PCM playback, phone-call/audio-focus interruption, background/resume, and a no-capability fallback. Assert that no production build can return the hard-coded phrase.

Exit Criteria: Android voice is genuinely functional and lifecycle-safe, or it is unavailable by design with an accessible deterministic fallback; no simulated speech path ships in release.

BUS-P0-07: Configure a genuine Android release identity and signing pipeline

Severity: P0 (Release Blocker)

Astra Gate / WCAG Mapping: Production security and supply-chain control; no direct WCAG criterion

Target Files & Symbols: `android/app/build.gradle.kts` `android.defaultConfig`, `buildTypes.release`; `android/app/src/main/AndroidManifest.xml`; CI/release secrets configuration

Observed Defect / Gap: The application ID is `com.example.helanapp` and the release build explicitly uses `signingConfigs.getByName("debug")`. A production artifact signed with the shared debug key cannot establish a secure update identity and is not a releasable package.

Required Production Solution: Freeze an organization-owned application ID; configure upload/release signing through environment-backed properties or Play App Signing; fail the release build when release credentials are absent; separate debug/staging/production product configuration and API endpoints; never commit keystore material.

Test Specification: Add CI jobs that build a signed release AAB, inspect package/application ID and certificate fingerprint, assert it differs from the debug certificate, and reject `debug` signing in a release variant.

Exit Criteria: A reproducible release AAB has the approved application ID and non-debug signing identity, with secrets outside source control.

P1 — Critical resilience and accessibility

BUS-P1-01: Add ordered, stale-aware live telemetry semantics

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 7 and 8; WCAG 2.2 SC 1.3.1 and 4.1.3

Target Files & Symbols: `lib/domain/transit/entities/bus_position.dart` `BusPosition`; `lib/data/models/transport_models.dart` `LiveBusMetrics`; `lib/data/datasources/live_bus_movement_engine.dart`; `lib/data/repositories/transport_repository.dart`; `live route/journey` widgets

Observed Defect / Gap: Telemetry contains a wall-clock timestamp but no source sequence/generation, freshness threshold, accuracy, or ordering guard. UI treats any snapshot as “LIVE GPS” and substitutes plausible default ETA/speed before data arrives. A delayed tick, clock rollback, reconnect, or old route stream can overwrite newer state and be announced as live.

Required Production Solution: Add `route/bus/session` IDs, monotonic sequence, UTC source/received timestamps, accuracy, and freshness state. Repository reducers must reject older generation/sequence values and derive `loading/live/stale/offline/error`; UI must never invent live values. Pause announcements while stale and expose last-updated time and recovery.

Test Specification: Add `test/data/live_telemetry_ordering_test.dart`: deliver ticks 10, 12, 11; duplicate 12; clock-skewed timestamps; route switch; 30/60/120-second silence; and reconnect generation. Assert state never regresses, stale/offline transitions occur once, and fake defaults are absent.

Exit Criteria: Out-of-order/old-route telemetry cannot update or announce current state; every live surface exposes deterministic freshness.

BUS-P1-02: Model stop occurrences for reverse, circular, and repeated-stop travel

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 1 and 8; WCAG 2.2 SC 3.3.2 Labels or Instructions

Target Files & Symbols: `lib/domain/transit/entities/transit_route.dart` `TransitRoute`. `stopsBetween`; `lib/data/models/transport_models.dart` `Route.stopsBetween`; `lib/data/repositories/transport_repository.dart` `findRoutes`; `lib/domain/ticketing/entities/fare_engine.dart`

Observed Defect / Gap: Both route models use `indexOf(stopId)` and return empty when the destination index is not greater than the origin. This cannot distinguish repeated physical stops, circular wrap-around, branches, or reverse travel on a single route. Empty results are indistinguishable from invalid/no journey and can feed incorrect fare/stop counts.

Required Production Solution: Represent ordered `StopOccurrence` values with occurrence ID, sequence, direction/pattern, and physical stop ID. Resolve a specific origin/destination occurrence pair through a trip pattern; support or explicitly reject wrap/reverse with a typed failure. Feed the resolved segment to both fare and journey logic.

Test Specification: Add `test/domain/route_segment_resolution_test.dart` with A-B-C-B-D, circular A-B-C-A, reverse travel, same occurrence, same physical stop at different occurrences, and branch patterns. Assert exact occurrence sequences, hop counts, typed ambiguity, and fare parity.

Exit Criteria: Segment resolution never relies on physical-stop `indexOf`; circular/repeated/reverse behavior is explicit and tested.

BUS-P1-03: Turn AnnouncementCoordinator into a route-scoped audio arbiter

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 6, 7, and 9; WCAG 2.2 SC 4.1.3 Status Messages

Target Files & Symbols: `lib/core/a11y/announcement_coordinator.dart`; `AnnouncementCoordinator`; `lib/features/route_details/route_details_page.dart`; `lib/features/tickets/ticket_booking_suite_page.dart`; assistant audio/controller files

Observed Defect / Gap: Visual status text is updated before route filtering, so an inactive route can overwrite visible status. Active route is set during widget build and is not reliably cleared. Distinct ETA messages bypass exact-string debounce and can speak every two seconds. High/normal messages are dropped while assistant audio plays; urgent messages collide with audio. Direction is forced LTR. Some screens call `SemanticsService.announce` directly during build, bypassing the coordinator; hidden `IndexedStack` steps remain mounted and can announce.

Required Production Solution: Use explicit route/screen registration tokens with dispose; filter before changing state; convert raw ticks into transition events; maintain bounded priority queues with expiry/deduplication; coordinate pause/duck/stop with TTS/PCM; derive locale/text direction; prohibit direct announcements outside the coordinator; suppress non-current routes and offstage steps.

Test Specification: Add `test/a11y/announcement_arbiter_test.dart`: two routes, hidden `IndexedStack` step, rapidly changing ETA, assistant PCM/TTS, urgent interruption, locale RTL, page disposal, and expired queued messages. Assert exact delivered sequence, no inactive visual update, no collision, and no duplicate output.

Exit Criteria: One audited coordinator owns every announcement; inactive/hidden content is silent; urgent and queued messages have deterministic audio behavior.

BUS-P1-04: Enforce modal semantics, focus restoration, and 48dp controls in the assistant overlay

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 2, 4, 5, and 9; WCAG 2.2 SC 2.4.3 Focus Order, 2.4.11 Focus Not Obscured, 2.5.8 Target Size

Target Files & Symbols: `lib/features/ai_assistant/floating_ai_assistant_overlay.dart` build, `_buildWindow`, `_buildVoiceControl`, `_buildPromptChips`; `lib/features/ai_assistant/floating_assistant_controller.dart`

Observed Defect / Gap: The background is visually dimmed with a `GestureDetector`, but underlying semantics are not blocked. The window has no modal traversal group or explicit initial/restored focus. Bubble repositioning is drag-only; no corner actions exist. Several controls use 38–40dp constraints or compact density despite a 48dp token. Infinite pulse/glow animations ignore reduced motion. The bubble can be clamped and notify listeners during build.

Required Production Solution: Wrap modal state in `BlockSemantics/ExcludeSemantics` as appropriate and a closed-loop traversal group; focus the window heading/primary control on open and restore the bubble on close. Add semantic actions/buttons for four corners and reset position. Enforce 48dp minimums without compact density; honor disable-animations/reduced-motion; move clamping to layout callbacks without notifying during build.

Test Specification: Extend overlay widget tests at 320dp and 300% scale with TalkBack semantics enabled. Assert underlying page nodes disappear in modal mode, focus remains inside, close restores bubble focus, all controls are at least 48x48, keyboard/switch users can reposition, and animations stop when requested.

Exit Criteria: Overlay interaction is fully operable without drag, cannot expose/obscure underlying focus, restores focus, and passes target/reduced-motion tests.

BUS-P1-05: Remove fixed-height and fixed-row failures at 200–300% text scale

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 3, 4, and 10; WCAG 2.2 SC 1.4.4 and 1.4.10

Target Files & Symbols: `lib/features/route_search/route_search_page.dart`; `_LocationSelector`, `_RouteCard`, `_StopPickerSheet`; `lib/core/a11y/map_text_alternative_widget.dart`; `lib/features/route_details/route_details_page.dart`; `lib/features/tickets/booking_checkout_dialog.dart`; `lib/features/tickets/ticket_booking_suite_page.dart`; safety/assistant cards

Observed Defect / Gap: Location selectors are fixed at 52dp, compact data is placed in non-wrapping rows, passenger/payment options force three cards in one row, many labels use ellipsis, and map alternative metrics assume horizontal space. At 320dp with 200–300% text these patterns clip, overflow, or hide decision-relevant text.

Required Production Solution: Replace fixed heights with minimum constraints, use `Wrap` or breakpoint-driven column layouts, allow labels and values to reflow without semantic loss, place secondary content below primary content, and use scrollable dialog bodies with pinned accessible actions. Avoid ellipsis for fares, stop names, errors, and state.

Test Specification: Create `test/a11y/reflow_matrix_test.dart` rendering all named surfaces at widths 320/360/600, portrait/landscape, text 1.0/2.0/3.0, and high contrast. Assert no overflow exceptions, clipped semantics, hidden buttons, or horizontal scrolling for ordinary cards/forms.

Exit Criteria: The complete search-to-ticket-to-journey flow remains readable and operable at 320dp and 300% platform text scale.

BUS-P1-06: Eliminate duplicate semantics and formalize focus contracts

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 4 and 5; WCAG 2.2 SC 1.3.1, 2.4.3, 2.4.6, and 4.1.2

Target Files & Symbols: `lib/core/a11y/map_text_alternative_widget.dart` build; route/ticket cards; checkout dialog and stop-picker sheets; safety confirmation dialogs; page headings

Observed Defect / Gap: Several compound widgets add a parent `Semantics(label:...)` while leaving identical child text exposed, causing duplicate readout. The map alternative uses the legacy boolean header property. Dialogs/sheets do not define explicit initial focus or restore the invoking control after asynchronous closure. Status and selection cards rely on prose labels instead of native selected/button/radio state.

Required Production Solution: For atomic summaries use `excludeSemantics:true`; otherwise remove the parent label and expose structured children. Use semantic heading levels, selected/toggled/value state, and standard controls. Create a navigation/focus policy: page heading on route entry, first error on failed submit, invoker restoration on dialog close, and no focus movement from data ticks.

Test Specification: Add semantics snapshots and focus traversal tests for every compound card, dialog, bottom sheet, and route transition. Assert each fact is spoken once, role/state are present, traversal matches task order, validation focuses the error, and dismiss restores the exact invoker.

Exit Criteria: Critical screens have approved semantics snapshots and deterministic initial/restored focus with no duplicated spoken content.

BUS-P1-07: Implement explicit OSRM, tile, and offline failure states

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 7, 8, and 11; WCAG 2.2 SC 3.3.1 and 3.3.3

Target Files & Symbols: `lib/data/services/osrm_routing_service.dart` `fetchRoute`; `lib/data/datasources/live_bus_movement_engine.dart` `_refreshPath`; `lib/features/journey/journey_page.dart` `_LiveMap`; `lib/features/route_details/route_details_page.dart`

Observed Defect / Gap: OSRM requests have no timeout, retry policy, response status check, payload-size limit, or typed failure. Every exception silently returns a straight line. Tile failures have no user-visible text state/retry. The route UI can label fallback/default values as live, preventing users from knowing that road routing or maps failed.

Required Production Solution: Return typed routing outcomes: road route, geometric fallback, timeout, offline, rate-limited, invalid response. Add bounded timeout and jittered retry only for transient failures; validate HTTP status/content and cancel obsolete requests. Expose a persistent text banner with last update, fallback meaning, and retry. Keep the list/text journey fully usable when tiles fail.

Test Specification: Add fake-client tests for connect timeout, slow body, 400/429/500, malformed/oversized JSON, route switch cancellation, offline startup, recovery, and tile errors. Assert bounded completion, correct fallback geometry, one plain-language status, working text journey, and no “live/road accurate” claim during fallback.

Exit Criteria: All network failure classes produce deterministic typed state and an accessible recovery action within a bounded time.

BUS-P1-08: Add voice permission recovery, reconnection backoff, and lifecycle control

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 6, 9, and 11; WCAG 2.2 SC 3.3.3 and 4.1.3

Target Files & Symbols: `lib/features/ai_assistant/gemini_live_session.dart` `connect`, `sendUserText`, `dispose`; `lib/features/ai_assistant/web_speech_real.dart`; `lib/features/ai_assistant/audio_speech_engine.dart`; `assistant controller/screen`

Observed Defect / Gap: WebSocket connection is one-shot: no reconnect state, exponential backoff, authentication/model capability validation, ping/idle handling, or retry cancellation. REST has no timeout. Browser global autoplay listeners are added to window and document and never removed. Microphone denial/permanent denial/background states are not modeled; recognition simply restarts. Assistant audio is not wired to `AnnouncementCoordinator.isAudioPlaying`.

Required Production Solution: Implement a connection state machine with capped exponential backoff+jitter, foreground/network gating, cancellation tokens, and explicit offline/manual retry. Add REST timeouts and structured errors. Model microphone permission states and settings recovery. Register/remove browser listeners through disposable handles. Integrate audio focus with the announcement arbiter and stop capture/playback on background or full-screen transitions. Validate the configured model through an integration contract rather than a hard-coded unverified name.

Test Specification: Use fake socket/HTTP/audio/permission adapters to test drop-before-setup, repeated close, 401/403/429/5xx, backoff cap, offline/online, app pause/resume, permanent denial, listener disposal, TalkBack announcement during PCM, and model rejection. Assert no reconnect storm, no capture in background, and actionable state text.

Exit Criteria: Voice sessions recover predictably, respect permissions/lifecycle, release all resources, and never collide with accessibility speech.

BUS-P1-09: Replace color claims with machine-verified semantic contrast

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gate 3; WCAG 2.2 SC 1.4.3 Contrast (Minimum), 1.4.11 Non-text Contrast

Target Files & Symbols: `lib/core/tokens/app_semantic_colors.dart`; `lib/core/theme/app_theme.dart`; hard-coded colors in route, ticket, safety, and assistant feature widgets

Observed Defect / Gap: The standard primary blue #0284C7 with white is about 4.10:1 and fails 4.5:1 normal-text AA. High-contrast error #FF3333 on #121212 is about 5.15:1, not AAA 7:1. Many feature widgets bypass semantic tokens with fixed light/dark colors, so theme/high-contrast mode cannot guarantee contrast.

Required Production Solution: Define foreground/background token pairs with documented minimum ratios; adjust failing pairs; distinguish text, large text, icon/border, focus, and disabled requirements. Replace feature hex values with semantic roles. Add non-color state cues and a visible high-contrast focus indicator. Rename the theme if every normal-text pair is not verified at 7:1.

Test Specification: Add `test/a11y/semantic_color_contrast_test.dart` that enumerates every allowed token pair in light/dark/high-contrast themes and asserts 4.5:1 normal text, 3:1 large/non-text, and 7:1 for every pair marketed AAA. Add a source lint against feature-level `Color(0x...)` except approved illustrations.

Exit Criteria: All rendered semantic pairs meet their declared ratio automatically; no feature bypasses tokens for functional content.

BUS-P1-10: Restore active journey, tickets, and settings after process death

Severity: P1 (Critical Resilience / A11y)

Astra Gate / WCAG Mapping: Astra Gates 2, 7, and 12; WCAG 2.2 SC 3.3.7 Redundant Entry

Target Files & Symbols: `lib/features/journey/journey_controller.dart`; `lib/features/tickets/ticket_controller.dart`; `lib/data/models/ticket_model.dart` `TicketRepository`; `lib/core/settings/app_settings_controller.dart`; `lib/data/datasources/local_json_store.dart`; routing/restoration setup

Observed Defect / Gap: Journey and tickets are in memory; process death loses an active journey and pass. Settings writes are fired without awaiting or surfacing errors, so rapid mutations can appear saved before disk completion. No restoration ID/state contract exists for route, selected stops, dialog progress, or assistant window.

Required Production Solution: Persist minimal versioned journey/ticket state transactionally; revalidate expiry and route availability on restore; never persist secrets or raw microphone/camera data. Expose

save failures. Use Flutter restoration for navigation/form state where appropriate and keep critical domain state in a durable repository. Await/serialize settings commits and flush on lifecycle pause.

Test Specification: Add restart tests that kill/recreate the app during route selection, active journey, ticket issuance, cancellation, settings changes, and assistant open state. Assert critical state restores or presents an explicit recovery path; expired/stale state is not revived; no secret API key is persisted.

Exit Criteria: Process death cannot silently erase a valid ticket or leave a journey in a false state; persistence failures are visible and recoverable.

P2 — Production polish and governance

BUS-P2-01: Minimize Android permissions and add contextual permission rationale

Severity: P2 (Production Polish)

Astra Gate / WCAG Mapping: Astra Gate 12; privacy and user-control requirement

Target Files & Symbols: `android/app/src/main/AndroidManifest.xml`; onboarding/settings permission flows; native feature adapters

Observed Defect / Gap: Fine/coarse location and microphone permissions are declared although the inspected Android voice path is simulated and location behavior is largely fixture-based. Broad declarations increase privacy surface without a matching native capability/rationale path.

Required Production Solution: Declare only permissions required by the production flavor; request just in time from the triggering action; explain purpose before the system prompt; support denial/permanent denial and settings recovery; document retention and sharing. Remove permissions from demo/offline flavors.

Test Specification: Add manifest merge tests per flavor and Android permission-flow tests for deny, deny-and-don't-ask, revoke while running, approximate location, and feature fallback.

Exit Criteria: Every manifest permission maps to a working, user-initiated feature and an accessible denial fallback; unused permissions are absent.

BUS-P2-02: Enable release shrinking and measure startup/resource cost

Severity: P2 (Production Polish)

Astra Gate / WCAG Mapping: Production performance; indirectly supports WCAG timing/stability

Target Files & Symbols: `android/app/build.gradle.kts` `buildTypes.release`; ProGuard/R8 rules; `lib/main.dart`; startup services

Observed Defect / Gap: Release minification/resource shrinking are not enabled. Multiple app-scoped controllers, overlays, JSON store initialization, and voice/map dependencies start near app boot without measured startup cost.

Required Production Solution: Enable R8 and resource shrinking for release, add only evidence-based keep rules, and run release smoke tests. Defer voice/map/session initialization until first use, preserve accessible first-frame content, and define cold/warm startup and memory budgets.

Test Specification: CI must build minified APK/AAB, launch every reflection/plugin-dependent feature, compare binary size, and collect macrobenchmark cold/warm startup plus memory after ten navigation cycles. Assert no accessibility semantics delay/regression.

Exit Criteria: Minified release passes smoke tests and documented startup/memory budgets without plugin or semantics breakage.

BUS-P2-03: Make CI the source of truth for tests, ADRs, and public claims

Severity: P2 (Production Polish)

Astra Gate / WCAG Mapping: All gates; governance and evidence integrity

Target Files & Symbols: README.md; docs/adr/ADR-001-phase1-domain-and-a11y-contracts.md; missing ADR-002 path; CI workflows; test inventory

Observed Defect / Gap: The request claims 236 tests and ADR-002, while public main reports 214 and the ADR path is unavailable. README also claims strict 48dp controls, uncapped scaling, high-contrast compliance, Android voice behavior, and authoritative fares that the inspected source does not consistently implement.

Required Production Solution: Generate test count/coverage/build badges from CI; publish ADR-002 or remove the reference; add claim-to-test traceability for all Astra gates; fail documentation checks when named files are absent or accessibility/release claims lack test evidence. Record the audited commit SHA in every release report.

Test Specification: Add a documentation verification job that validates linked files, runs analysis/tests/a11y matrices, derives counts, scans release signing/configuration, and produces an immutable evidence artifact.

Exit Criteria: README, ADRs, release notes, and CI artifacts agree automatically; no manual test count or unsupported compliance claim remains.

BUS-P2-04: Strengthen static analysis and modernize browser interop

Severity: P2 (Production Polish)

Astra Gate / WCAG Mapping: Astra Gates 2 and 9; maintainability prerequisite

Target Files & Symbols: analysis_options.yaml; lib/features/ai_assistant/web_speech_real.dart; all async UI callbacks

Observed Defect / Gap: Only default recommended lints are enabled. Browser speech uses `dart:js` and `eval`; global listener lifecycle is implicit. The codebase contains build-time side effects, unawaited persistence, and async UI callbacks that require stronger enforcement.

Required Production Solution: Enable strict casts/inference/raw types plus lints for discarded futures, context after async gaps, close sinks, cancel subscriptions, and deprecated APIs. Replace `eval` and legacy JS interop with typed `dart:js_interop/web` bindings; hold removable listener references; introduce narrowly scoped suppressions with rationale.

Test Specification: Run `flutteranalyze--fatal-infos--fatal-warnings` in CI; add browser tests under a restrictive CSP that forbids `eval` and verify initialization/disposal across repeated app mounts.

Exit Criteria: Strict analysis is clean, browser voice works without `eval`, and all listeners/futures have explicit ownership.

3 Section C: Master Implementation Order for the Coding Agent

The coding agent should use the following zero-regression loop. At every numbered task: create or update the specified tests first, implement only that task, run domain/unit/widget tests plus static analysis, and commit only after the baseline is green.

1. **Freeze evidence.** Record the current commit SHA, capture current test/analysis output, publish or correct ADR-002, and make CI report its own test count (BUS-P2-03). This prevents further audit drift.
2. **Correct text scaling first.** Implement BUS-P0-03 and its 320dp/300% harness. All subsequent UI work must pass that harness.
3. **Unify currency/domain flow.** Implement BUS-P0-01 before modifying ticket state, because the state machine must persist the canonical quote.
4. **Build ticket ledger/state machine.** Implement BUS-P0-02, including durable storage and QR/payment feature gating. Run migration/restart/replay tests.
5. **Make teardown deterministic.** Implement BUS-P0-04; use the resulting cancellation and ownership primitives for later network/audio work.
6. **Correct assistant command truthfulness.** Implement BUS-P0-05 before reconnect logic so retries cannot duplicate emergency or financial commands.
7. **Implement/gate native Android audio.** Complete BUS-P0-06, then run accessibility audio tests on a real Android device.
8. **Fix release identity/signing.** Complete BUS-P0-07 and create a signed internal release build. No public release work proceeds until all P0 tests pass together.
9. **Strengthen route and telemetry domain.** Implement BUS-P1-02 followed by BUS-P1-01; migrate journey/fare/UI consumers and run end-to-end route tests.
10. **Build the central audio/announcement arbiter.** Implement BUS-P1-03, then wire assistant audio through it as part of BUS-P1-08.
11. **Harden overlay and focus.** Implement BUS-P1-04 and BUS-P1-06 together; validate with keyboard, switch-style traversal, TalkBack, and modal focus restoration.
12. **Complete responsive reflow.** Implement BUS-P1-05 screen by screen using the established reflow matrix; do not add per-screen scaler clamps.
13. **Correct colors and token adoption.** Implement BUS-P1-09 after layouts stabilize; regenerate light/dark/high-contrast goldens and contrast reports.
14. **Add deterministic network degradation.** Implement BUS-P1-07, then finish WebSocket/permission/lifecycle resilience in BUS-P1-08 using the same typed state/failure patterns.
15. **Add process restoration.** Implement BUS-P1-10 after ticket, journey, network, and lifecycle state machines are stable.
16. **Reduce platform exposure.** Complete BUS-P2-01, removing permissions not justified by the now-final native feature set.
17. **Modernize and tighten analysis.** Complete BUS-P2-04, fixing all strict-analysis findings rather than broadly suppressing them.

18. **Optimize the release build.** Complete BUS-P2-02 with R8/resource shrinking, release smoke tests, startup metrics, and memory-leak navigation loops.
19. **Final verification.** Run the full test suite; fatal static analysis; 320dp at 100/200/300% text; light/dark/high-contrast contrast tests; Android TalkBack core journeys; keyboard/switch traversal; offline/429/5xx recovery; process-death restoration; assistant audio collision; emergency confirmation; signed minified AAB install/upgrade; and manual validation with blind/low-vision participants where ethically feasible. Publish the generated evidence artifact and update the readiness score only from those results.

Release decision

Decision: NO-GO for production release. The application may continue as a clearly labeled prototype. A production release requires all seven P0 tasks and P1 tasks BUS-P1-01 through BUS-P1-10 to be complete, with independently rerun CI and device evidence. The most consequential defects are not cosmetic: the live ticket flow bypasses the currency authority, Android voice is simulated, an AI emergency tool can report an action that did not occur, platform text scaling is altered, and the Android release is debug-signed.

Primary inspected sources

- Public repository: <https://github.com/sricharan24-byte/HELEN> (main, retrieved 20 September 2026).
- Key inspected sources include `lib/main.dart`, domain transit/ticketing entities and repositories, service locator, live movement engine, OSRM service, semantic tokens/theme, announcement/map alternatives, route/journey/ticket/safety/assistant features, `pubspec.yaml`, `analysis_options.yaml`, Android manifest/build configuration, ADR-001, README, and available tests.
- Flutter accessibility guidance and test guidelines: <https://docs.flutter.dev/ui/accessibility> and https://api.flutter.dev/flutter/flutter_test/AccessibilityGuideline-class.html.
- W3C WCAG 2.2 for non-web software guidance: <https://www.w3.org/TR/wcag2ict-22/>.
- Android release signing and shrinking guidance: <https://developer.android.com/studio/publish/app-signing> and <https://developer.android.com/build/shrink-code>.